METTLER TOLEDO

Hydrofluoric Acid Content of Lithium-ion Battery Electrolyte

Potentiometric Titration in the Mettler Toledo Titration Excellence T5 To Determine HF Content

Lithium-ion batteries (LIB) are commonly used for portable electronic devices and electric vehicles. They are growing in popularity because of their high energy density. The lithium-ion battery stores energy that is released by an electrochemical reaction between the anode and the cathode material. Both the anode and the cathode contain positively charged lithium ions. During discharge, the oxidation reaction at the anode produces electrons and lithium cations. The electrons flow through an external wire to the cathode. To complete the electric circuit, lithium cations, flow though the electrolyte to the cathode where they recombine with the electron in a reduction reaction. Although the lithium ions formally do not partake in the electrochemical reaction, it is important that they are efficiently carried to the respective electrode by the electrolyte.

Keywords or phrases: Lithium- ion, electrolyte, hydrofluoric acid, titration, battery, lithium, Mettler Toledo

Introduction

The electrolyte is usually composed of the electrolyte salt and a number of additives (e.g. organic carbonates). Lithium hexafluorophosphate $LiPF_6$ is the most frequently used electrolyte salt in LIB electrolytes. It dissolves well in organic carbonates, is electrochemically stable and helps to conduct lithium cations. However, the use of $LiPF_6$ also brings up a problem. It is not stable against hydrolysis. Traces of water in the electrolyte lead to the formation of hydrofluoric acid HF, which is considered as battery poison destroying the electrodes and lowering the capacity. Therefore, the amount of HF in the lithium-ion battery electrolyte should be as low as possible. The HF content serves as a quality indicator and is listed in the specification of the electrolyte.

This application note describes a method to determine the hydrofluoric acid content in a lithium-ion battery electrolyte by an aqueous acid-base titration with potentiometric indication.

Sample Preparation and Procedures

Add approximately 40 mL of deionized water into a 100 mL titration beaker and place the beaker on the balance.

Tare the balance and add ca. 3 g of the electrolyte using a syringe.

Then place the beaker on the InMotion autosampler rack and start method as shown in Table 2.

Chemistry

$HF + NaOH \rightleftharpoons NaF + H_2O$

Chemicals

Titrant: Aq. sodium hydroxide, NaOH, c(NaOH) = 0.01 mol/L.

Sample: Lithium hexafluorophosphate solution in ethylene carbonate and ethyl methyl carbonate, 1.0 M, $LiPF_6$ in EC/EMC=50/50 (v/v), battery grade

Analyte: Hydrofluoric acid, HF, M = 20.01 g/mol, z = 1. Auxiliary Reagents: deionized water.

Standard: Potassium hydrogen phthalate, $C_8H_5KO_4$, M = 204.22 g/mol, z = 1

Instruments and Accessories

- Titration Excellence T5
- InMotion flex 100 mL
- Burette DV1010 10 mL
- Compact Stirrer
- DGi111-SC or DGi115-SC electrode
- Excellence balance XPR603S Delta Range

Results

	Sample Size (g)	R1: Content (ppm)
1	2.8542	75.02
2	3.1214	76.78
3	2.897	78.62
4	2.7795	75.61
5	2.8144	76.52
6	3.2041	76.97
Mean		**76.59**
s		1.24
srel		**1.62%**

The results show a good repeatability with a relative standard deviation of 1.7%. The results are listed in chronological order. The inexistence of a trend towards increasing values indicates that the hydrolysis during the sample handling is negligible.

Measured Values

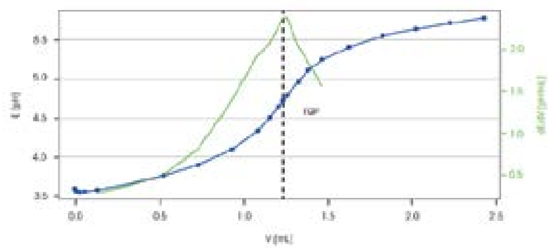

Figure 1: Titration curve (blue) and the first derivative of the titration curve (green) of sample 6/6.

Method

Sample	
Sample Type	Sample
Number of IDs	1
ID 1	Electrolyte
Entry type	Weight
Lower Limit	0
Upper Limit	5 g
Density	1 g/mL
Correction Factor	1
Temperature	25.0 °C
Titrator reader	None
Number of sample factors	0
Titration stand	
Type	InMotion T/Tower A
Titration Stand	InMotion T/1A
Head Position	Sample
Stir	
Speed	30%
Duration	5 s
Titration (EQP)1	
Titrant	
Tirant	NaOH
Concentration	0.01 mol/L
Sensor	
Type	pH
Sensor	DGi111-SC
Unit	mV
Temperature Acquisition	
Temperature Measurement	No
Stir	
Speed	30%
Predispense	
Mode	None
Wait time	0 s
Control	
Control	Cautious
Mode	Acid/Base
Titrant Addition	Dynamic
dE (set value)	10 mV
dV (min)	0.005 mL
dV (max)	0.2 mL
Mode	Equilibrium controlled
dE (set value)	0.5 mV
dt	2 s
t (min)	3 s
t (max)	30 s

Evaluation and Recognition	
Procedure	Standard
Threshold	1.5 pH/mL
Tendency	Positive
Ranges	0
Add. EQP criteria	No
Termination	
At Vmax	5 mL
At potential	No
At slope	No
After number of recognised EQPs	Yes
Number of EQPs	1
Combined Termination Criteria	No
Condition	No
Calculation R1	
Result	Content
Result Unit	ppm
Formula	R1=Q*C/m
Constant C	C=M*1000/z
M	M[Hydrofluoric acid]
Decimal Places	3
Result Limits	No
Extra Statistical Functions	No
Send to Buffer	No
Write to Smart Tag	None
Condition	No
Rinse	
Auxiliary Reagent	Water
Rinse Cycles	1
Vol. per cycle	10 mL
Position	Current Position
Drain	No
Condition	No
End of Sample	
Park	
Titration Stand	InMotion T/1A
Position	Condition Beaker
Condition	No
Record	
Report Template	Titration Report

Table 2: Experimental Method

t (s)	V (mL)	E (pH)	dE/dV (mV/mL)
0	0	3.586	n/a
3	0.005	3.553	n/a
6	0.01	3.55	n/a
9	0.0225	3.549	n/a
12	0.0535	3.554	n/a
17	0.131	3.58	0.3
23	**0.325**	**3.66**	**0.38**
30	0.525	3.766	0.51
37	**0.725**	**390.10%**	**0.82**
47	0.925	4.097	1.38
60	1.083	4.343	1.93
72	1.153	4.505	2.07
86	1.204	4.646	2.25
EQP	**1.23252**	**4.726**	**2.37**
100	1.2555	4.7	2.37
118	1.318	4.964	2.05
138	**1.3805**	**5.113**	**1.86**
159	1.4645	5.25	1.86
181	**1.623**	**540.80%**	**n/a**
206	**1.823**	**555.50%**	**n/a**
222	2.023	5.643	n/a
238	2.223	5.723	n/a
251	2.423	5.783	n/a

Conclusion

$LiPF_6$ hydrolyses slowly in aqueous solutions, while it is hydrolysing faster in organic solutions with traces of water[1]. Thus, the titration is conveniently performed in water using aqueous NaOH as titrant.

To prevent the further formation hydrofluoric acid by hydrolysis, the sample should be titrated readily.

This method was only tested on the presented sample. The compositions of Lithium-ion battery electrolytes vary, and some electrolytes may contain additives that hydrolyse to form of hydrofluoric acid very quickly. For such samples, it may be necessary to strictly work in non-aqueous conditions.

The titre of aq. 0.01 M sodium hydroxide solution is determined using potassium hydrogen phthalate (KHP) as the primary standard. As the optimum amount of standard per measurement is very low (5 mg), a 5 mg/mL aq. KHP standard solution is prepared.

After every sample, the electrode, stirrer and tubing are rinsed with deionized water.

The titrations are performed in plastic beakers as HF is able to etch glass.

Waste Disposal and Safety Measures

Although the hydrofluoric acid content of the electrolyte is quite low, every contact with skin or eyes should be prevented. Use gloves, safety goggles and a lab coat. Work in a fume hood. After the titration, all solutions are neutralized and disposed as aqueous waste.

Reference

1. Stich M., Göttlinger M., Kurniawan M., Schmidt U., Bund A., J. Phys. Chem., 2018, 122, 16, 8836-8842.